Week 10: Advanced SQL Query System Scenario:

1.Find products whose price is greater than the average product price.

```
SQL> SELECT PRODUCT_NAME,PRICE FROM PRODUCT WHERE PRICE > (SELECT AVG(PRICE) FROM PRODUCT);

PRODUCT_NAME              PRICE
-------------------- ----------
Laptop                    55000
Smart TV                  45000
Mobile                    25000
Refrigerator              38000
Washing Machine           32000
Air Conditioner           42000
Tablet                    28000
Camera                    35000
Gaming Laptop             85000
Dishwasher                45000

10 rows selected.
```

2. Find Products Above Average Price Task:

- **Find products priced above the average price.**

```
SQL> SELECT PRODUCT_NAME,PRICE,CATEGORY_ID FROM PRODUCT WHERE PRICE > (SELECT AVG(PRICE) FROM PRODUCT);

PRODUCT_NAME              PRICE CATEGORY_ID
-------------------- ---------- -----------
Laptop                    55000           1
Smart TV                  45000           1
Mobile                    25000           2
Refrigerator              38000           3
Washing Machine           32000           3
Air Conditioner           42000           3
Tablet                    28000           1
Camera                    35000           1
Gaming Laptop             85000           1
Dishwasher                45000           3

10 rows selected.
```

- **Display product name, category, and price.**

```
SQL> SELECT PRODUCT_NAME,CATEGORY_ID,PRICE FROM PRODUCT;

PRODUCT_NAME         CATEGORY_ID      PRICE
-------------------- ----------- ----------
Laptop                         1      55000
Smart TV                       1      45000
Mobile                         2      25000
Keyboard                       2       1500
Mouse                          2        800
Headphones                     2       2500
Refrigerator                   3      38000
Washing Machine                3      32000
Air Conditioner                3      42000
Tablet                         1      28000
Printer                        2      12000

PRODUCT_NAME         CATEGORY_ID      PRICE
-------------------- ----------- ----------
Smart Watch                    2       5000
Microwave                      3      15000
Camera                         1      35000
Speaker                        2       4000
Monitor                        1      18000
Power Bank                     2       1800
Gaming Laptop                  1      85000
Dishwasher                     3      45000
Electric Kettle                3       2200

20 rows selected.
```

- **Identify expensive products in each category.**

```
SQL> SELECT CATEGORY_ID,MAX(PRICE) FROM PRODUCT GROUP BY CATEGORY_ID;

CATEGORY_ID MAX(PRICE)
----------- ----------
          1      85000
          2      25000
          3      45000
```

- **Find products with prices higher than a selected category average.**

```
SQL> SELECT * FROM PRODUCT WHERE PRICE > (SELECT AVG(PRICE) FROM PRODUCT WHERE CATEGORY_ID = 1);

PRODUCT_ID PRODUCT_NAME         CATEGORY_ID      PRICE STOCK_QUANTITY
---------- -------------------- ----------- ---------- --------------
       101 Laptop                         1      55000             10
       102 Smart TV                       1      45000             15
       118 Gaming Laptop                  1      85000              5
       119 Dishwasher                     3      45000              6
```

3.Identify Customers with Maximum Purchases Task:

- **Customer who spent the highest amount.**

```
SQL> SELECT CUSTOMER_ID , SUM(TOTAL_AMOUNT) FROM ORDERS GROUP BY CUSTOMER_ID HAVING SUM(TOTAL_AMOUNT) = (SELECT MAX(SUM(TOTAL_AMOUNT))
FROM ORDERS GROUP BY CUSTOMER_ID);

CUSTOMER_ID SUM(TOTAL_AMOUNT)
----------- -----------------
          7              5000
```

- **Customers with maximum number of orders.**

```
SQL> SELECT CUSTOMER_ID,COUNT(ORDER_ID) FROM ORDERS GROUP BY CUSTOMER_ID HAVING COUNT(ORDER_ID) = (SELECT MAX(COUNT(ORDER_ID)) FROM OR
DERS GROUP BY CUSTOMER_ID);

CUSTOMER_ID COUNT(ORDER_ID)
----------- ---------------
          1               1
          6               1
          2               1
          4               1
          5               1
          8               1
          3               1
          7               1
          9               1
         10               1

10 rows selected.
```

- **Customers whose spending is above average.**

```
SQL> SELECT CUSTOMER_ID,SUM(TOTAL_AMOUNT) FROM ORDERS GROUP BY CUSTOMER_ID HAVING SUM(TOTAL_AMOUNT) > (SELECT AVG(TOTAL_AMOUNT) FROM O
RDERS);

CUSTOMER_ID SUM(TOTAL_AMOUNT)
----------- -----------------
          1              2800
          4              4000
          3              3000
          7              5000
```

- **Top 5 valuable customers.**

```
SQL> SELECT CUSTOMER_ID,TOTAL_SPENDING FROM (SELECT CUSTOMER_ID,SUM(TOTAL_AMOUNT) AS TOTAL_SPENDING FROM ORDERS GROUP
BY CUSTOMER_ID ORDER BY SUM(TOTAL_AMOUNT) DESC) WHERE ROWNUM <= 5;

CUSTOMER_ID TOTAL_SPENDING
----------- --------------
          7           5000
          4           4000
          3           3000
          1           2800
          2           1500
```

4.Generate Complex Business Queries

Query 1: Best-Selling Product

- **Display product name.**
- **Total quantity sold.**
- **Total revenue generated.**

```
SQL> SELECT P.PRODUCT_NAME,
  2         SUM(OD.QUANTITY) AS TOTAL_QUANTITY,
  3         SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  4  FROM PRODUCT P
  5  JOIN ORDER_DETAILS OD
  6  ON P.PRODUCT_ID = OD.PRODUCT_ID
  7  GROUP BY P.PRODUCT_NAME
  8  ORDER BY TOTAL_QUANTITY DESC;

PRODUCT_NAME         TOTAL_QUANTITY TOTAL_REVENUE
-------------------- -------------- -------------
Smart TV                         12         15030
Laptop                           11        605000
```

Query 2: High-Value Customers

Find customers whose total purchase amount is greater than average customer spending.

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.FIRST_NAME,
  3         C.LAST_NAME,
  4         SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME
  9  HAVING SUM(O.TOTAL_AMOUNT) > (
 10      SELECT AVG(TOTAL_SPENDING)
 11      FROM (
 12          SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SPENDING
 13          FROM ORDERS
 14          GROUP BY CUSTOMER_ID
 15      )
 16  );

CUSTOMER_ID FIRST_NAME                                         LAST_NAME                                          TOTAL_SPENDING
----------- -------------------------------------------------- -------------------------------------------------- --------------
          1 Arun                                               Kumar                                                        2800
          3 Karthik                                            Raj                                                          3000
          4 Swetha                                             Mohan                                                        4000
          7 Rohan                                              Sharma                                                       5000
```

Query 3: Category Performance Analysis Find:

- **Highest revenue-generating category.**

```
SQL> SELECT *
  2  FROM (
  3      SELECT
  4          P.CATEGORY_ID,
  5          SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  6      FROM PRODUCT P
  7      JOIN ORDER_DETAILS OD
  8      ON P.PRODUCT_ID = OD.PRODUCT_ID
  9      GROUP BY P.CATEGORY_ID
 10      ORDER BY TOTAL_REVENUE DESC
 11  )
 12  WHERE ROWNUM = 1;

CATEGORY_ID TOTAL_REVENUE
----------- -------------
          1        620030
```

- **Number of products sold in each category.**

```
SQL> SELECT
  2      P.CATEGORY_ID,
  3      SUM(OD.QUANTITY) AS TOTAL_PRODUCTS_SOLD
  4  FROM PRODUCT P
  5  JOIN ORDER_DETAILS OD
  6  ON P.PRODUCT_ID = OD.PRODUCT_ID
  7  GROUP BY P.CATEGORY_ID;

CATEGORY_ID TOTAL_PRODUCTS_SOLD
----------- -------------------
          1                  23
```

- **Average product price by category.**

```
SQL> SELECT C.CATEGORY_NAME,
  2         AVG(P.PRICE) AS AVERAGE_PRICE
  3  FROM PRODUCT P
  4  JOIN CATEGORY C
  5  ON P.CATEGORY_ID = C.CATEGORY_ID
  6  GROUP BY C.CATEGORY_NAME;

CATEGORY_NAME                                      AVERAGE_PRICE
-------------------------------------------------- -------------
Haircare                                             29033.3333
Makeup                                                     6575
Skincare                                             44333.3333
```

Query 4: Customer Purchase History Display:

- **Customer name**
- **Number of orders**
- **Total purchase amount**

```
SQL> SELECT C.FIRST_NAME,
  2         C.LAST_NAME,
  3         COUNT(O.ORDER_ID) AS TOTAL_ORDERS,
  4         SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME;

FIRST_NAME                                         LAST_NAME                                          TOTAL_ORDERS TOTAL_PURCHASE
-------------------------------------------------- -------------------------------------------------- ------------ --------------
Arun                                               Kumar                                                     1           2800
Karthik                                            Raj                                                       1           3000
Pooja                                              Nair                                                      1            800
Harini                                             Krishnan                                                  1            500
Keerthi                                            Suresh                                                    1           1200
Sanjay                                             Das                                                       1            300
Swetha                                             Mohan                                                     1           4000
Imran                                              Ali                                                       1           1000
Rohan                                              Sharma                                                    1           5000
Meena                                              Ravi                                                      1           1500

10 rows selected.
```

- **Most purchased product**

```
SQL> SELECT CUSTOMER_ID, PRODUCT_NAME, TOTAL_QUANTITY
  2  FROM (
  3      SELECT O.CUSTOMER_ID,
  4             P.PRODUCT_NAME,
  5             SUM(OD.QUANTITY) AS TOTAL_QUANTITY,
  6             ROW_NUMBER() OVER (
  7                 PARTITION BY O.CUSTOMER_ID
  8                 ORDER BY SUM(OD.QUANTITY) DESC
  9             ) AS RN
 10      FROM ORDERS O
 11      JOIN ORDER_DETAILS OD
 12      ON O.ORDER_ID = OD.ORDER_ID
 13      JOIN PRODUCT P
 14      ON OD.PRODUCT_ID = P.PRODUCT_ID
 15      GROUP BY O.CUSTOMER_ID, P.PRODUCT_NAME
 16  )
 17  WHERE RN = 1;

CUSTOMER_ID PRODUCT_NAME         TOTAL_QUANTITY
----------- -------------------- --------------
          2 Smart TV                         12
```

5. Advanced SQL Report

Report 1: Premium Product Report

Display:

- **Product Name**
- **Category**

- **Price**
- **Stock Availability Condition**

```
SQL> SELECT PRODUCT_NAME,
  2         CATEGORY_ID,
  3         PRICE,
  4         CASE
  5             WHEN STOCK_QUANTITY > 0 THEN 'AVAILABLE'
  6             ELSE 'OUT OF STOCK'
  7         END AS STOCK_CONDITION
  8  FROM PRODUCT
  9  WHERE PRICE > (
 10      SELECT AVG(PRICE)
 11      FROM PRODUCT
 12  );

PRODUCT_NAME         CATEGORY_ID      PRICE STOCK_CONDIT
-------------------- ----------- ---------- ------------
Laptop                         1      55000 AVAILABLE
Smart TV                       1      45000 AVAILABLE
Mobile                         2      25000 AVAILABLE
Refrigerator                   3      38000 AVAILABLE
Washing Machine                3      32000 AVAILABLE
Air Conditioner                3      42000 AVAILABLE
Tablet                         1      28000 AVAILABLE
Camera                         1      35000 AVAILABLE
Gaming Laptop                  1      85000 AVAILABLE
Dishwasher                     3      45000 AVAILABLE

10 rows selected.
```

Report 2: Customer Value Report

Display:

- **Customer Name**
- **Total Orders**
- **Total Spending**
- **Customer Category**

```
SQL> SELECT C.FIRST_NAME,
  2         C.LAST_NAME,
  3         COUNT(O.ORDER_ID) AS TOTAL_ORDERS,
  4         SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING,
  5         CASE
  6             WHEN SUM(O.TOTAL_AMOUNT) > 100000
  7             THEN 'PREMIUM'
  8             ELSE 'REGULAR'
  9         END AS CUSTOMER_CATEGORY
 10  FROM CUSTOMER C
 11  JOIN ORDERS O
 12  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 13  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME;

FIRST_NAME                                         LAST_NAME                                          TOTAL_ORDERS TOTAL_SPENDING CUSTOME
-------------------------------------------------- -------------------------------------------------- ------------ -------------- -------
Arun                                               Kumar                                                     1           2800 REGULAR
Karthik                                            Raj                                                       1           3000 REGULAR
Pooja                                              Nair                                                      1            800 REGULAR
Harini                                             Krishnan                                                  1            500 REGULAR
Keerthi                                            Suresh                                                    1           1200 REGULAR
Sanjay                                             Das                                                       1            300 REGULAR
Swetha                                             Mohan                                                     1           4000 REGULAR
Imran                                              Ali                                                       1           1000 REGULAR
Rohan                                              Sharma                                                    1           5000 REGULAR
Meena                                              Ravi                                                      1           1500 REGULAR

10 rows selected.
```

Report 3: Sales Performance Report

A. Product-wise sales

```
SQL> SELECT P.PRODUCT_NAME,
  2         SUM(OD.QUANTITY) AS TOTAL_QUANTITY,
  3         SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_SALES
  4  FROM PRODUCT P
  5  JOIN ORDER_DETAILS OD
  6  ON P.PRODUCT_ID = OD.PRODUCT_ID
  7  GROUP BY P.PRODUCT_NAME;

PRODUCT_NAME         TOTAL_QUANTITY TOTAL_SALES
-------------------- -------------- -----------
Laptop                           11      605000
Smart TV                         12       15030
```

B.Category-wise revenue

```
SQL> SELECT P.CATEGORY_ID,
  2         SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  3  FROM PRODUCT P
  4  JOIN ORDER_DETAILS OD
  5  ON P.PRODUCT_ID = OD.PRODUCT_ID
  6  GROUP BY P.CATEGORY_ID;

CATEGORY_ID TOTAL_REVENUE
----------- -------------
          1        620030
```

C. Top-selling products

```
SQL> SELECT P.CATEGORY_ID,
  2         SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  3  FROM PRODUCT P
  4  JOIN ORDER_DETAILS OD
  5  ON P.PRODUCT_ID = OD.PRODUCT_ID
  6  GROUP BY P.CATEGORY_ID;

CATEGORY_ID TOTAL_REVENUE
----------- -------------
          1        620030
```

Report 4: Business Decision Report

Identify:

A.High-performing products

```
SQL> SELECT P.PRODUCT_NAME,
  2         SUM(OD.QUANTITY) AS TOTAL_SALES
  3  FROM PRODUCT P
  4  JOIN ORDER_DETAILS OD
  5  ON P.PRODUCT_ID = OD.PRODUCT_ID
  6  GROUP BY P.PRODUCT_NAME
  7  HAVING SUM(OD.QUANTITY) > (
  8      SELECT AVG(TOTAL_QUANTITY)
  9      FROM (
 10          SELECT SUM(QUANTITY) AS TOTAL_QUANTITY
 11          FROM ORDER_DETAILS
 12          GROUP BY PRODUCT_ID
 13      )
 14  );

PRODUCT_NAME         TOTAL_SALES
-------------------- -----------
Smart TV                      12
```

B. Low-performing products

```
SQL> SELECT P.PRODUCT_NAME,
  2         NVL(SUM(OD.QUANTITY), 0) AS TOTAL_SALES
  3  FROM PRODUCT P
  4  LEFT JOIN ORDER_DETAILS OD
  5  ON P.PRODUCT_ID = OD.PRODUCT_ID
  6  GROUP BY P.PRODUCT_NAME
  7  ORDER BY TOTAL_SALES ASC;

PRODUCT_NAME         TOTAL_SALES
-------------------- -----------
Camera                         0
Monitor                        0
Smart Watch                    0
Mobile                         0
Tablet                         0
Power Bank                     0
Printer                        0
Air Conditioner                0
Mouse                          0
Headphones                     0
Speaker                        0
Keyboard                       0
Electric Kettle                0
Gaming Laptop                  0
Dishwasher                     0
Microwave                      0
Washing Machine                0
Refrigerator                   0
Laptop                        11
Smart TV                      12

20 rows selected.
```

C. High-value customers

```
SQL> SELECT CUSTOMER_ID,
  2         SUM(TOTAL_AMOUNT) AS TOTAL_SPENDING
  3  FROM ORDERS
  4  GROUP BY CUSTOMER_ID
  5  HAVING SUM(TOTAL_AMOUNT) > (
  6      SELECT AVG(TOTAL_SPENDING)
  7      FROM (
  8          SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SPENDING
  9          FROM ORDERS
 10          GROUP BY CUSTOMER_ID
 11      )
 12  );

CUSTOMER_ID TOTAL_SPENDING
----------- --------------
          1           2800
          4           4000
          3           3000
          7           5000
```

D. Revenue trends (month-wise)

```
SQL> SELECT TO_CHAR(ORDER_DATE, 'MM-YYYY') AS MONTH,
  2         SUM(TOTAL_AMOUNT) AS TOTAL_REVENUE
  3  FROM ORDERS
  4  GROUP BY TO_CHAR(ORDER_DATE, 'MM-YYYY')
  5  ORDER BY MONTH;

MONTH   TOTAL_REVENUE
------- -------------
09-2026          4300
10-2026         15300
11-2026           500
```